

Week 2 Summary – Data Structures and Functions

Objective

The objective of Week 2 was to understand Python data structures and functions and apply them to solve programming problems and perform basic data cleaning tasks.

Concepts Learned

- Learned about Python data structures such as **Lists, Tuples, Dictionaries, and Sets**.
- Understood how to create and use **Functions** for reusable code.
- Explored **Lambda Functions** for writing short anonymous functions.
- Learned **Recursion** and how a function can call itself.
- Used **List Comprehension** to create and manipulate lists efficiently.

Hands-on Programs Completed

1. Count the frequency of elements using a dictionary
2. Sort a list using a lambda function

Client Project

Project Title: Student Data Cleaner

Project Description

Developed a Python program to clean student data by accepting a list of student names, removing duplicate entries, sorting the names alphabetically, and searching for a specific student in the cleaned list.

Skills Used

- Lists
- Sets
- Dictionaries
- Functions
- Loops
- Conditional Statements
- Data Cleaning Techniques

Learning Outcome

This week improved my understanding of Python data structures and functions. I learned how to organize, clean, and process data efficiently using lists, dictionaries, and sets. I also gained

practical experience in writing reusable functions, using lambda expressions, recursion, and list comprehensions. These concepts provide a strong foundation for working with datasets in Data Science.